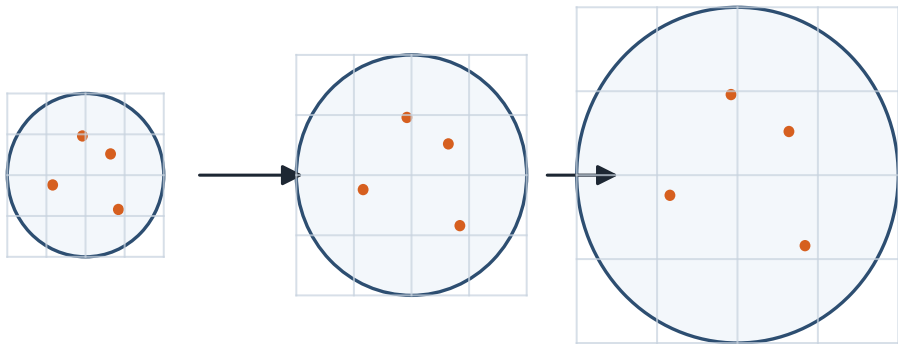


physical separations obey  $D_{\text{phys}}(t) = a(t) D_{\text{com}}$



early:  $a(t) < 1$

today:  $a_0 = 1$

later:  $a(t) > 1$

photon wavelengths stretch with the same factor:  $1 + z = a_0/a(t_{\text{em}})$